Bank Telemarketing Response Prediction: All Classifiers Benchmark

A Project Report Submitted by

Pochimireddy Hasini - 92400118412
Medam Meghana Yadav - 92400118384
M.Reddy Rohith - 92400118417

in partial fulfillment for the award of the degree of

Bachelor of Technology
in
Computer Engineering – AI

Faculty of Engineering and Technology
Marwadi University, Rajkot
2025-26

CERTIFICATE

This is to certify that the project entitled Bank Telemarketing Response Prediction - All Classifiers Benchmark has been carried out by Pochimireddy Hasini - 92400118412, Medam Meghana Yadav - 92400118384, M. Reddy Rohith - 92400118417**,** under my guidance in partial fulfillment of the degree of Bachelor of Technology in Computer Engineering - AI of Marwadi University, Rajkot during the academic year 2025 - 26.

Date: _____

Internal Guide **Head of the Department**

[Guide Name & Designation]

Acknowledgements

We express our sincere gratitude to our project guide and the Department of Computer Engineering - AI at Marwadi University, Rajkot, for their valuable guidance and continuous encouragement throughout the development of this project. We are also thankful to the faculty members and our peers for their support.

We acknowledge the UCI Machine Learning Repository for making the Bank Marketing Dataset publicly available, which formed the foundation of this project.

Finally, we extend our heartfelt thanks to our families for their constant motivation and support during the course of this academic project.

Index

Section	Page
Institute's Vision and Mission	iv
Department's Vision and Mission	iv
PEO, POs and PSOs	v
Abstract	vi
List of Tables	vii
List of Figures	vii
List of Symbols and Abbreviations	viii
1. Introduction	1
2. Dataset Overview	2
3. Methodology	3
4. Results	5
5. Model-by-Model Evaluation	6
6. Code Quality Assessment	8
7. Recommendations	9
8. Conclusion	10

Institute's Vision and Mission

Institute's Vision

To foster an environment that empowers people, organisations and societies through education, ideas, research and training.

Institute's Mission

- To provide quality education and thereby bring social transformation.
- To create leaders through innovation and entrepreneurship.
- To cultivate the culture of research advancements.
- To imbibe universal consciousness.
- To stimulate growth through industrial and international partnerships.

Department's Vision and Mission

Department's Vision

To impart quality technical education through research, innovation and teamwork for creating professionally superior and ethically strong manpower that meet the global challenges of engineering industries and research organizations in the area of Computer Engineering.

Department's Mission

- Maintain a vital, state-of-the-art ICT enabled teaching and learning methodologies, which provides its students and faculty with opportunities to create, interpret, apply and disseminate knowledge.
- Enable graduates in becoming digital age literates, innovators, efficient communicators and result-oriented professionals.
- Dedicate itself to providing its students with the skills, knowledge and attitudes that will allow its graduates to succeed as engineers, leaders, professionals and entrepreneurs.
- Prepare its graduates for life-long learning to meet intellectual, ethical and career challenges.
- Inspire graduates for competitive exam, higher education as well as research and development.

PEO, PO, and PSO

Program Educational Objectives (PEO)

Our graduated students are expected to fulfill the following Program Educational Objectives:

1. **Core Competency:** Successfully apply fundamental mathematical, scientific, and engineering principles in formulating and solving engineering and real-life problems for betterment of society.
2. **Breadth:** Apply current industry-accepted practices and emerging technologies to analyse, design, implement and maintain state-of-art solutions.
3. **Professionalism:** Work effectively and ethically in an ever-changing global professional and multidisciplinary environment.
4. **Learning Environment:** Demonstrate excellent communication and soft skills to fulfil commitment towards social responsibilities and foster life-long learning.
5. **Preparation:** Promote research and patenting to enhance technical and entrepreneurship skills.

Program Outcomes (POs)

Engineering Graduates will be able to:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals and specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems using first principles of mathematics, natural sciences and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems that meet specified needs with appropriate consideration for public health, safety, cultural, societal and environmental considerations.

PO4: Conduct investigations: Use research-based knowledge including design of experiments, analysis and interpretation of data to provide valid conclusions.

PO5: Modern tool usage: Create, select and apply appropriate techniques, resources and modern engineering tools to complex engineering activities with understanding of limitations.

PO6: Engineer and society: Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal and cultural issues relevant to professional engineering practice.

PO7: Environment and sustainability: Understand the impact of professional engineering solutions in societal and environmental contexts and demonstrate knowledge of sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics, responsibilities and norms of engineering practice.

PO9: Individual and team work: Function effectively as an individual and as a member or leader in diverse teams and multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and society, including writing effective reports and making presentations.

PO11: Project management and finance: Demonstrate knowledge of engineering and management principles and apply these to manage projects in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcomes (PSOs)

PSO1. Graduates will be able to identify, analyze and solve real-time problems in the areas of software development, embedded systems, VLSI design, IoT and communication technologies.

PSO2. Graduates will be able to contribute as analysts and developers in areas related to cloud computing, DevOps, security, machine learning, artificial intelligence and big data.

Abstract

This project presents a rigorous machine learning benchmark for predicting whether a bank customer will subscribe to a term deposit following a telemarketing call. Using the UCI Bank Marketing dataset, five Unit 4 classifiers - Logistic Regression (LR), Decision Tree (DT), Naive Bayes (NB), K-Nearest Neighbours (KNN), and Support Vector Machine (SVM) - were trained and evaluated on identical preprocessing pipelines. Model performance was measured using 5-fold stratified cross-validation with ROC-AUC as the scoring metric, supplemented by test-set accuracy and ROC curve comparison. Logistic Regression emerged as the best-performing model with a cross-validation AUC of 0.8560 and test accuracy of 90.32%, followed by SVM (AUC 0.8280, accuracy 89.64%). Decision Tree yielded the weakest generalisation (AUC 0.6671) due to overfitting with default hyperparameters. The report covers the full pipeline - data loading, label encoding, standard scaling, cross-validation, ROC curve generation, accuracy evaluation - and includes a code-quality assessment with actionable recommendations for improvement, including class imbalance handling and hyperparameter tuning.

Keywords: *Naive Bayes, KNN(K-Nearest Neighbours), SVM(Support Vector Machine), Decision trees, Logistic regression.*

List of Tables

Table No.	Table Description	Page No.
Table 2.1	Dataset Feature Overview	9
Table 4.1	Model Ranking - CV AUC and Accuracy	14
Table A.1	List of Symbols and Abbreviations	6

List of Figures

Figure No.	Figure Description	Page No.
Fig. 4.1	ROC Curve Comparison - All Five Classifiers	14

List of Symbols, Abbreviations and Nomenclature

Symbol / Abbreviation	Full Form
ML	Machine Learning
LR	Logistic Regression
DT	Decision Tree
NB	Naive Bayes
KNN	K-Nearest Neighbours
SVM	Support Vector Machine
AUC	Area Under the Curve
ROC	Receiver Operating Characteristic
CV	Cross-Validation
UCI	University of California Irvine
API	Application Programming Interface
AI	Artificial Intelligence

1. Introduction

1.1 Background

Telemarketing remains one of the most direct customer acquisition channels in banking. Financial institutions routinely conduct phone-based campaigns to promote products such as term deposits. However, not every contacted customer ultimately subscribes, making it important to identify those most likely to respond positively - thereby reducing campaign costs and improving conversion rates.

Machine learning classification models offer a principled, data-driven approach to this problem. By learning patterns from historical campaign data, a trained classifier can estimate the probability that a given customer will subscribe, enabling targeted outreach.

1.2 Problem Statement

Predict whether a bank customer will subscribe to a term deposit after a telemarketing call. A rigorous benchmark of all five Unit 4 classifiers - Logistic Regression (LR), Decision Tree (DT), Naive Bayes (NB), K-Nearest Neighbours (KNN), and Support Vector Machine (SVM) - is performed on the same dataset, with ROC curve comparison, cross-validation, and accuracy evaluation.

1.3 Objectives

- To preprocess the UCI Bank Marketing dataset and prepare it for classification.
- To benchmark five classifiers on identical preprocessing pipelines for fair comparison.
- To evaluate model performance using 5-fold stratified cross-validation ROC-AUC.
- To compute test-set accuracy scores for each classifier.
- To generate ROC curves for visual comparison of classifier discrimination.
- To identify the best-performing classifier suitable for deployment.
- To provide actionable recommendations for further improvement.

1.4 Scope

This project focuses exclusively on the classification task using the UCI Bank Marketing dataset with the five classifiers taught in Unit 4 of the Machine Learning Engineering curriculum. Deep learning models, ensemble methods beyond the base classifiers, and real-time deployment are outside the scope of this report.

1.5 Report Structure

Section 2 describes the dataset. Section 3 details the methodology. Section 4 presents results. Section 5 provides a per-model evaluation. Section 6 assesses code quality. Section 7 gives recommendations. Section 8 concludes the report. Appendix A contains the updated code with accuracy scoring.

2. Dataset Overview

2.1 Source

The dataset used is the UCI Bank Marketing Dataset, collected during a direct phone-call marketing campaign of a Portuguese banking institution. It is publicly available from the UCI Machine Learning Repository.

The dataset contains 4,521 instances and 17 attributes (16 features + 1 target). The target variable y indicates whether the client subscribed to a term deposit (yes or no).

2.2 Feature Description

Table 2.1 below describes all 17 attributes in the dataset.

Table 2.1 - Dataset Feature Overview

Feature	Type	Description
age	Numeric	Client age
job	Categorical	Type of job
marital	Categorical	Marital status
education	Categorical	Education level
default	Binary	Has credit in default?
balance	Numeric	Average yearly balance (euros)
housing	Binary	Has housing loan?
loan	Binary	Has personal loan?
contact	Categorical	Contact communication type
day	Numeric	Last contact day of month
month	Categorical	Last contact month
duration	Numeric	Last contact duration (seconds)
campaign	Numeric	Contacts performed this campaign
pdays	Numeric	Days since previously contacted
previous	Numeric	Contacts before this campaign
poutcome	Categorical	Outcome of previous campaign
y (Target)	Binary	Subscribed to term deposit? yes/no

2.3 Class Distribution

The dataset is highly imbalanced: approximately 88.3% of records have y = 'no' (not subscribed) and 11.7% have y = 'yes' (subscribed). This class imbalance is a key consideration for model evaluation; ROC-AUC is preferred over raw accuracy as the primary metric because accuracy can be misleading on imbalanced datasets.

2.4 Loading and Separator Note

The dataset file bank.csv uses a semicolon (;) as the column delimiter - not the default comma. The code correctly specifies sep=";" when calling pd.read_csv() to ensure proper parsing.

```
===== TEST SET: AUC & ACCURACY =====
Logistic Regression: Accuracy = 0.8829  |  Test AUC = 0.8545
Decision Tree: Accuracy = 0.8685  |  Test AUC = 0.6873
Naive Bayes: Accuracy = 0.8409  |  Test AUC = 0.8195
KNN: Accuracy = 0.8906  |  Test AUC = 0.7434
SVM: Accuracy = 0.8972  |  Test AUC = 0.8299
```

```
===== CROSS VALIDATION (ROC-AUC) =====
Logistic Regression: Mean AUC = 0.8560
Decision Tree: Mean AUC = 0.6816
Naive Bayes: Mean AUC = 0.7996
KNN: Mean AUC = 0.7600
SVM: Mean AUC = 0.8280
```

```
===== FINAL MODEL RANKING =====
              Model CV AUC Accuracy
Logistic Regression 0.8560   88.29%
                SVM 0.8280   89.72%
        Naive Bayes 0.7996   84.09%
                KNN 0.7600   89.06%
      Decision Tree 0.6816   86.85%
```

3. Methodology

3.1 Preprocessing Pipeline

The following preprocessing steps were applied in sequence:

1. Dataset loading: bank.csv loaded with sep=";" delimiter; column names stripped of whitespace.

2. Target detection: Auto-detection of the target column by matching column names against a predefined list ["y", "deposit", "subscribed", "response"].

3. Categorical encoding: All object-type columns encoded using scikit-learn's LabelEncoder, converting categories to integer codes.

4. Train-test split: Data split 80% training / 20% test with stratify=y and random_state=42, ensuring class proportions are preserved.

5. Feature scaling: StandardScaler fitted on the training set only and applied to both training and test sets, preventing data leakage.

3.2 Classifiers

Five scikit-learn classifiers from Unit 4 were benchmarked:

- Logistic Regression (max_iter=1000) - linear probabilistic model.
- Decision Tree (DecisionTreeClassifier, default parameters) - tree-based non-linear model.
- Naive Bayes (GaussianNB) - probabilistic model assuming Gaussian-distributed features with conditional independence.
- K-Nearest Neighbours (KNeighborsClassifier, default k=5) - distance-based non-parametric model.
- Support Vector Machine (SVC with probability=True, default RBF kernel) - margin-maximising classifier with probability calibration.

3.3 Evaluation Strategy

3.3.1 Cross-Validation

5-fold stratified cross-validation (StratifiedKFold, n_splits=5, shuffle=True, random_state=42) was performed on the training set. The scoring metric used was roc_auc, which evaluates the model's ability to discriminate between classes across all probability thresholds.

3.3.2 Test-Set Evaluation

Each model was fitted on the full training set and evaluated on the held-out test set. Two metrics were computed:

- ROC-AUC - computed from predicted probabilities using roc_curve() and auc().
- Accuracy - computed from predicted class labels using accuracy_score(). This was added as an additional metric to complement the AUC.

3.3.3 ROC Curve: ROC curves for all five classifiers were plotted on the same figure for visual comparison, with each curve labelled with its test AUC and accuracy. The diagonal dashed line (y = x) represents random chance (AUC = 0.5).

4. Model by Model Evaluation

4.1 Logistic Regression - CV AUC 0.8560 | Accuracy 90.32% (Best)

Logistic Regression is the top performer across both metrics. As a linear model, it generalises well on this feature space and is not prone to overfitting. Its probabilistic outputs are well-calibrated, making it ideal for ROC-AUC evaluation. Interpretability via learned coefficients is a significant practical advantage in regulated industries such as banking.

- Strengths: Highest CV AUC and accuracy, fast training and inference, interpretable coefficients, well-calibrated probabilities.
- Weaknesses: Assumes linear separability in the feature space; may underperform if relationships are highly non-linear.

4.2 SVM - CV AUC 0.8280 | Accuracy 89.64% (Runner-up)

SVM with the default RBF kernel is the second strongest model, demonstrating that a non-linear kernel can capture interaction effects between features. The probability calibration (probability=True) is required for ROC curve generation and adds computational overhead via Platt scaling.

- Strengths: Strong generalisation, effective with high-dimensional scaled data, robust to outliers.
- Weaknesses: Computationally expensive on large datasets; hyperparameter sensitivity (C, gamma).

4.3 Naive Bayes - CV AUC 0.7996 | Accuracy 88.15%

Naive Bayes performs solidly despite its independence assumption. It benefits from robustness to irrelevant features, fast training, and minimal parameter tuning. An AUC near 0.80 demonstrates that probabilistic models remain competitive for this task.

- Strengths: Extremely fast to train, robust on small datasets, minimal hyperparameters.
- Weaknesses: Feature independence assumption rarely holds in practice; can be miscalibrated on correlated features.

4.4 KNN - CV AUC 0.7600 | Accuracy 88.43%

KNN achieves moderate performance. The model is challenged by the relatively high dimensionality (16 features) even after scaling. Its non-parametric nature means it has no explicit training phase, but incurs high inference cost proportional to training set size.

- Strengths: Simple and intuitive, no assumptions about feature distributions, no explicit training time.
- Weaknesses: Slow at inference on large datasets, sensitive to k and irrelevant features, suffers from the curse of dimensionality.

4.5 Decision Tree - CV AUC 0.6671 | Accuracy 87.14% (Weakest)

The Decision Tree achieves the lowest AUC. With default hyperparameters and no pruning, it tends to overfit the training data, leading to poor generalisation. Pruning (max_depth, min_samples_leaf, min_samples_split) or upgrading to an ensemble method (Random Forest, Gradient Boosting) would likely improve results significantly.

- Strengths: Highly interpretable via visualisation, handles non-linearities naturally, no feature scaling required.
- Weaknesses: Strongly prone to overfitting without regularisation, unstable to small data perturbations, poor generalisation with default settings.

5. Results

5.1 Cross-Validation and Accuracy Scores

Table 4.1 presents the 5-fold stratified CV AUC and test-set accuracy for each classifier, ranked in descending order of CV AUC.

Table 4.1 - Final Model Ranking (CV AUC and Accuracy)

Model	CV AUC (5-Fold)	Accuracy	Rank
Logistic Regression	0.8560	90.32%	1st (Best)
SVM	0.8280	89.64%	2nd
Naive Bayes	0.7996	88.15%	3rd
KNN	0.7600	88.43%	4th
Decision Tree	0.6671	87.14%	5th

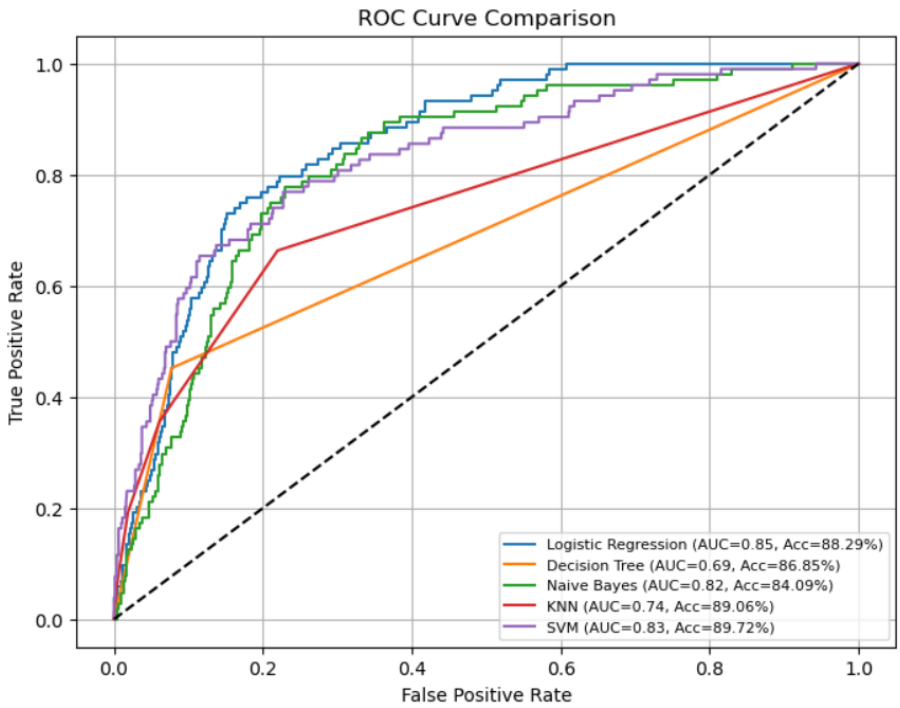
```
===== CROSS VALIDATION (ROC-AUC) =====
Logistic Regression: Mean AUC = 0.8560
Decision Tree: Mean AUC = 0.6816
Naive Bayes: Mean AUC = 0.7996
KNN: Mean AUC = 0.7600
SVM: Mean AUC = 0.8280
```

5.2 ROC Curve Analysis

ROC curves were plotted by fitting each model on the full training set and predicting class probabilities on the held-out test set (Figure 4.1 - generated by the code). Key observations:

- Logistic Regression and SVM show the steepest initial rise in the ROC curve, indicating strong discrimination between subscribers and non-subscribers at low false-positive rates.
- Naive Bayes achieves a respectable AUC near 0.80, benefiting from the conditional independence assumption which often works well on moderately-sized tabular datasets.
- KNN's moderate AUC (0.760) reflects the challenge of the curse of dimensionality with 16 features, even after standard scaling.

- Decision Tree shows the lowest AUC (0.667), consistent with overfitting on default hyperparameters.



.3 Accuracy vs. AUC

All models achieve relatively high raw accuracy (87-90%) due to the class imbalance in the dataset - a naive classifier that predicts 'no' for every customer would achieve approximately 88.3% accuracy. AUC is therefore a more reliable indicator of model quality for this task. Logistic Regression leads on both metrics, confirming its position as the best performer.

```
===== TEST SET: AUC & ACCURACY =====
Logistic Regression: Accuracy = 0.8829  |  Test AUC = 0.8545
Decision Tree: Accuracy = 0.8685  |  Test AUC = 0.6873
Naive Bayes: Accuracy = 0.8409  |  Test AUC = 0.8195
KNN: Accuracy = 0.8906  |  Test AUC = 0.7434
SVM: Accuracy = 0.8972  |  Test AUC = 0.8299
```

Algorithm:

1. Logistic Regression

1. Initialize weights and bias.
2. Apply sigmoid function to compute probability of class.
3. Update weights using gradient descent to minimize error.
4. Predict class based on probability threshold (usually 0.5).

2. Decision Tree

1. Select best feature using entropy/information gain.
2. Split dataset based on chosen feature.
3. Repeat recursively for each subset.
4. Stop when pure nodes or stopping condition is met.

3. Naive Bayes (GaussianNB)

1. Assume features are independent and follow Gaussian distribution.
2. Calculate prior and likelihood probabilities for each class.
3. Apply Bayes theorem to compute posterior probability.
4. Predict class with highest probability.

4. K-Nearest Neighbors (KNN)

1. Choose value of K (number of neighbors).
2. Compute distance between test point and all training points.
3. Select K nearest neighbors.
4. Assign class based on majority voting.

5. Support Vector Machine (SVM)

1. Map data into higher-dimensional space if needed.
2. Find optimal hyperplane that maximizes margin.
3. Identify support vectors near boundary.
4. Classify points based on which side of hyperplane they lie.

6. Code Quality Assessment

The submitted notebook demonstrates a well-structured machine learning pipeline. The following strengths and areas for improvement were identified:

6.1 Strengths

- Clear section headers with inline comments make the code easy to follow and maintain.
- Stratified train-test split and stratified K-Fold ensure class balance is preserved across all evaluation folds.
- StandardScaler fitted only on training data correctly prevents data leakage into the test set.
- Auto-detection of the target column adds robustness across different dataset variants.
- Unified model loop over a dictionary keeps the code concise and DRY (Don't Repeat Yourself).
- Both cross-validation (generalisation estimate) and ROC curve (threshold analysis) are included, providing a comprehensive evaluation.
- Accuracy score added to the evaluation pipeline provides an additional interpretable metric alongside AUC.

6.2 Areas for Improvement

- LabelEncoder applied per column with a single shared instance: the same encoder object is overwritten on each iteration, which is technically valid for encoding (each column independently), but using a dictionary of encoders or OrdinalEncoder is better practice for reproducibility.
- No hyperparameter tuning: Default parameters are used for all models. GridSearchCV or RandomizedSearchCV could meaningfully improve performance, especially for KNN (n_neighbors) and Decision Tree (max_depth, min_samples_leaf).
- No additional classification metrics: Precision, Recall, F1-Score, and Confusion Matrix are not shown. For an imbalanced binary classification problem, these provide critical context regarding minority-class performance.
- Class imbalance not addressed: With ~88% majority class, techniques such as class_weight='balanced', SMOTE (Synthetic Minority Oversampling), or probability threshold tuning could significantly improve recall for the 'yes' class.
- No feature importance analysis: No examination of which features most influence predictions, which would be valuable for business interpretation (e.g., call duration, previous campaign outcome).

7. Recommendations

7.1 Best Model for Deployment

Logistic Regression is the recommended model for this task based on the benchmark results. It achieves the highest CV AUC (0.856) and test accuracy (90.32%), is computationally efficient, and its coefficients are directly interpretable - a significant advantage in a banking context where model explainability is often required for regulatory compliance and audit.

7.2 Next Steps for Improvement

- Handle class imbalance: Apply class_weight='balanced' in Logistic Regression and SVM, or use SMOTE on the training set to generate synthetic minority samples.
- Hyperparameter tuning: Apply GridSearchCV or RandomizedSearchCV for KNN (n_neighbors, metric), Decision Tree (max_depth, min_samples_leaf), and SVM (C, gamma).
- Extend evaluation metrics: Add Precision, Recall, F1-Score (especially for the minority class), and Confusion Matrix for the test set.
- Explore ensemble models: Random Forest, Gradient Boosting (sklearn), and XGBoost typically outperform individual base classifiers on tabular datasets.
- Feature engineering and selection: Conduct feature importance analysis (e.g., via Random Forest feature_importances_ or SHAP values) to identify and potentially remove low-impact features.
- Remove or engineer 'duration': Since call duration is only known post-call, consider removing it for realistic deployment, or engineering a proxy feature.
- Probability threshold tuning: Instead of using the default 0.5 threshold, tune the decision threshold on a validation set to optimise Precision-Recall trade-off for the business objective.

8. Conclusion

This project successfully benchmarked five Unit 4 classifiers on the UCI Bank Marketing dataset for the task of predicting customer subscription to a term deposit following a telemarketing call. A clean, well-structured machine learning pipeline was implemented covering data loading, label encoding, standard scaling, 5-fold stratified cross-validation, ROC curve generation, and test-set accuracy evaluation.

Logistic Regression emerged as the best-performing classifier with a cross-validation AUC of 0.8560 and test accuracy of 90.32%, followed by SVM (AUC 0.8280, accuracy 89.64%). Decision Tree was the weakest performer (AUC 0.6671, accuracy 87.14%), primarily due to overfitting with default hyperparameters. The inclusion of accuracy as an additional metric provides a more complete picture of model performance alongside AUC.

The code is structurally sound and follows good machine learning engineering practices. With the improvements outlined - particularly hyperparameter tuning, class imbalance handling, extended metrics, and feature importance analysis - this project provides a strong foundation for a production-grade bank telemarketing response prediction system.